

UNIFYING SPEECH AND LANGUAGE THROUGH CONTINUOUS DUAL-FUSION FOR MEDIUM-RESOURCE LANGUAGES: THE CASES OF HUNGARIAN AND GREEK

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ABSTRACT

In medium-resource languages like Hungarian and Greek, Automatic Speech Recognition (ASR) suffers from scarce paired acoustic data, whereas abundant text data can be leveraged via Large Language Model (LLM) priors. To compensate for this acoustic deficit, we propose a Dual-Fusion architecture that conditions ASR decoding on continuous acoustic information at both the prompt level and inside the LLM’s intermediate layers. Our model bridges a frozen acoustic encoder with pre-trained LLMs via both prompt-level embeddings and deep, continuous-space alignment across multiple intermediate decoder layers. To actively control the semantic-acoustic trade-off, our methodology introduces asymmetric downsampling to preserve fine phonetic boundaries, multi-layer gated cross-attention, attention-weighted layer-wise fusion of the acoustic encoder’s hidden states, and target text perturbation. Evaluated on diverse Hungarian and Greek datasets, our framework reduces Word Error Rate (WER) compared to both standalone acoustic decoders and acoustic-linguistic models like Vox-KriKri, demonstrating consistent performance improvements across both languages.

Index Terms— Automatic Speech Recognition, Speech LLMs, Continuous Fusion, Medium-Resource ASR, Multimodal Alignment

1. INTRODUCTION

Automatic Speech Recognition (ASR) has advanced significantly with robust end-to-end models like Whisper [1] and data-efficient architectures like Canary [2], achieving state-of-the-art multilingual performance. However, these systems often degrade on medium-resource, morphologically complex languages such as Hungarian and Greek. We hypothesize this drop stems not solely from acoustic feature extraction, but partly from orthographic and morphological errors caused by a lack of deep, language-specific linguistic priors, consistent with evidence that external language models can improve low-resource ASR [3].

To bridge this gap, recent research couples pre-trained acoustic encoders with Large Language Models (LLMs) in Speech-Language Models (SLMs) [4], broadly following two integration strategies [5]: prompt-level approaches that prepend compressed acoustic embeddings as pseudo-text tokens, and cross-attention approaches that let text states attend to acoustic representations at intermediate layers. Early prompt-level strategies minimized architectural complexity: the SLAM-ASR framework [6] showed that a simple, trainable linear projector connecting a frozen encoder like Whisper to a frozen LLM yields competitive ASR. SALMONN [7] extended this with a learned Q-Former to bridge audio embeddings into the LLM space. We hypothesize that the rigid temporal pooling used by such prompt-only connectors risks destroying micro-phonetic boundaries required

for highly inflected languages; consistent with this concern, SLAM-ASR degrades under domain shift and acoustic perturbations [8].

For deeper semantic integration, cross-attention fusion was explored. The Whispering LLaMA framework [9] pioneered cross-modal generative error correction by integrating Whisper’s audio embeddings directly into the LLM via cross-attention adapters. A comparative connector study [10] reports that multi-head cross-attention, despite higher parameter cost than linear projectors, can yield better representational alignment than simple connectors. Furthermore, Whisper-Flamingo [11] established the efficacy of gated cross-attention to interleave frozen text blocks with continuous multimodal inputs, ensuring stable generation. In the speech domain, VoxKriKri [12] operates fully within a continuous latent space, fusing Whisper’s hidden decoder states with an LLM’s textual representations at a single selected layer, showing that intermediate layer injection yields robust cross-modal alignment for Greek.

A related question is which encoder representations to expose. Rather than passing only the final hidden state, SUPERB [13] established that different layers of a self-supervised speech encoder carry different phonetic and semantic information; follow-up work [14] shows learned weighted combination over all encoder layers can outperform single-layer extraction, and more recent work replaces static weights with input-dependent attention, e.g. Adaptive Layer Attention [15] and TASLA [16]. Finally, fused acoustic-LLM decoders often suffer the “lazy LLM” problem, ignoring acoustic information in favor of the text prior. We adopt a training-time token-substitution technique to mitigate this by weakening the model’s access to clean linguistic context during teacher forcing; this is related to, but distinct from, inference-time textual perturbation used to counter hallucination in multimodal LLMs [17].

Building on these findings, we propose a Dual-Fusion architecture that couples a frozen Whisper encoder with language-specialized LLM backbones for Hungarian and Greek, combining: (i) a *Dual-Fusion* strategy feeding acoustic information at both the prompt level and via cross-attention, with decoupled downsampling factors; (ii) *multiple injection layers* with *layer-wise fusion* of all Whisper encoder layers rather than only the final state; (iii) *gated cross-attention* scaling injected context with a learnable tanh gate; and (iv) *text perturbation*, substituting random tokens in the teacher-forced target sequence. We are not aware of a prior system that combines these within a single continuous, audio-conditioned architecture; we study whether doing so improves performance for both Hungarian and Greek, two typologically distinct medium-resource languages, and report a component-wise ablation (Section 4.5). Evaluated on multiple Hungarian and Greek corpora against standalone Whisper and VoxKriKri, our Dual-Fusion models yield consistent Word Error Rate reductions across both languages.

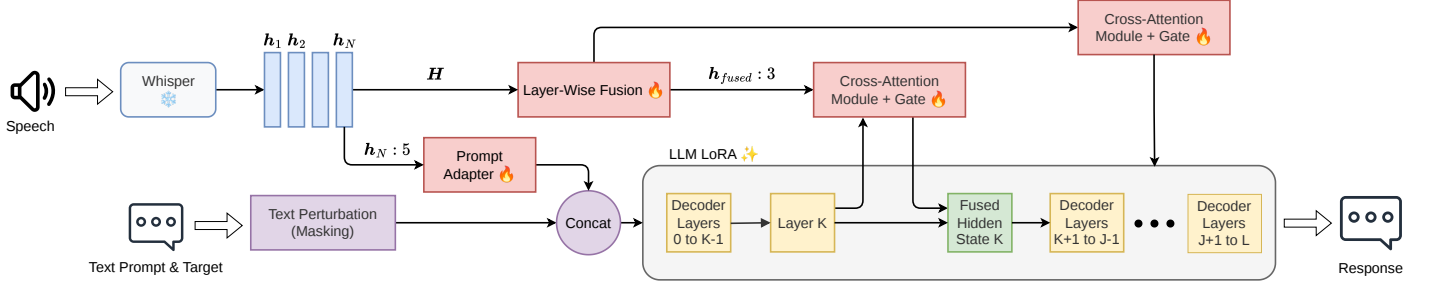


Fig. 1: Proposed Continuous Dual-Fusion ASR architecture: The speech instances are first processed by a frozen Whisper acoustic encoder, producing the acoustic hidden states $H = \{h_1 \dots h_N\}$. These states are processed in parallel by two distinct fusion pathways to bridge them into the LLM embedding space. In the prompt-level path, the final acoustic state (h_N) with dimensions $T \times D$ is downsampled by a factor of $k = 5$ and projected into the LLM space as prefix tokens of dimension $\lfloor T/5 \rfloor \times D_{LLM}$ via an MLP adapter. Concurrently, in the cross-attention path, all encoder states H undergo Layer-Wise Fusion and Conv1D downsampling by a finer factor of $k = 3$ to form a phonetically rich acoustic representations of dimension $\lfloor T/3 \rfloor \times D$. The LoRA-adapted LLM decoder follows; it processes the concatenated acoustic and text tokens, while aggregating the continuous acoustic representations via Gated Cross-Attention modules at multiple intermediate layers.

2. METHODOLOGY

The methodology couples a frozen Whisper encoder with a language-specific LLM (Racka-4B for Hungarian, Llama-Krikri-8B-Instruct for Greek) through prompt-level conditioning and intermediate cross-attention. An overview is given in Fig. 1.

2.1. Dual Fusion Strategy

To bridge the temporal mismatch between high-frequency acoustic frames and dense semantic tokens, we deploy a Dual-Fusion Strategy with asymmetric downsampling, integrating acoustic embeddings at two points: the prompt level and the intermediate cross-attention layers. In the *prompt path*, the final Whisper encoder state h_N is reshaped with a downsampling factor of 5 (yielding a 100ms resolution) and passed through a two-layer MLP $\text{Proj}(\cdot)$, then concatenated into the LLM’s input embedding sequence as prefix-like speech tokens:

$$\text{embed}(\text{prefix}) \parallel \text{Proj}(h_N : 5) \parallel \text{embed}(\text{instr}) \parallel \text{embed}(\text{target})$$

Crucially, this coarser downsampling factor prevents quadratic computational bloat in the LLM’s global self-attention sequence, while aligning with syllabic boundaries. Conversely, the *cross-attention path* injects speech-derived representations at selected layers using a finer Conv1D downsampler ($k = 3$, 60ms resolution). This preserves the micro-phonetic boundaries essential for highly inflected languages—specifically, rapid Hungarian agglutinative suffixes and unstressed Greek vowel transitions. Maintaining this denser resolution is computationally viable because it strictly localizes to cross-attention operations without expanding the LLM’s global sequence length. Ablations (Section 4.5) confirm $k = 3$ balances this model’s performance against the prohibitive cost of $k = 1$.

2.2. Multiple Injection Layers as Persistent Acoustic Memory

VoxKrikri injects at a single selected depth. We instead inject cross-attention simultaneously after several intermediate decoder blocks (e.g., after blocks [10, 20, 30] of 32 for KriKri, and [15, 25, 35] of 36 for Racka), with a fixed stride of ten blocks to sample early-, mid-, and late-decoder representations, since a single early exposure risks dilution as the hidden state propagates deeper. Repeated injection

maintains a persistent acoustic memory, re-grounding generation instead of drifting toward pure language-model continuation.

2.3. Layer-Wise Fusion of Acoustic Representations

VoxKrikri fuses only Whisper’s final hidden state; we instead use all encoder hidden states $\{h_0 h_1 \dots h_N\}$, since lower states preserve phonetic detail and higher states carry semantic abstraction. A *LayerWiseAttention* module mean-pools each state’s representation over the time dimension and scores it with a shared MLP, producing an *utterance-level* combination:

$$h_{fused}^{(j)} = \sum_{l=0}^N \alpha_l^{(j)} h_l, \quad \alpha_l^{(j)} = \text{softmax}_l \left(v^\top \tanh(W \bar{h}_l) \right)$$

2.4. Gated Cross-Attention

Injecting acoustic context at multiple layers along two pathways risks letting acoustic signal overwhelm the LLM’s own generative flow, especially for purely syntactic tokens (punctuation, function words) that should be governed by linguistic, not acoustic, evidence. Following the zero-initialized gating principle from Flamingo [18], each injection layer scales its cross-attention output with a single learnable scalar gate $g^{(j)}$:

$$\tilde{h}_{LLM} = h_{LLM} + \tanh \left(g^{(j)} \right) C_{audio}$$

2.5. Text Perturbation

To counter the “lazy LM” problem, where the decoder over-relies on the teacher-forced text prior and ignores the injected acoustic information, we introduce a dynamic text perturbation mechanism during training. For each sequence, eligible target-history tokens are independently replaced by the tokenizer’s padding token with a probability of $p = 10\%$. By stochastically masking the textual history during teacher forcing, the decoder is compelled to actively attend to the continuous acoustic representations to accurately predict the subsequent token. We set $p = 10\%$ per the ablation in Section 4.5, which minimizes validation WER relative to $p=0\%$ (+1.1), $p=20\%$ (+0.7), and $p=30\%$ (+1.1).

3. EXPERIMENTAL SETUP

3.1. Backbones and Injection Configuration

Both language configurations adopt the frozen Whisper-large-v3 [1] encoder (637M parameters). For Greek, the LLM backbone is Llama-KriKri-8B-Instruct [19] (8.2B base parameters, hidden dimension $D_{LLM} = 4096$, 32 decoder blocks); for Hungarian, elte-nlp/Racka-4B [20] (4.0B base parameters, $D_{LLM} = 2560$, 36 decoder blocks). The trainable bridge network dynamically scales to match each LLM’s hidden dimension space. Specifically, the bridge consists of a reshape adapter (19M for Hungarian, 21.5M for Greek), a shared Conv1D downsampler (5M for both), cross-attention modules (59M for Hungarian, 132M for Greek), and a layer-fusion module (1M for both). This brings the total trainable parameters in Stage 1 to 84M for Hungarian and 160M for Greek, approximately 1.78% of the full model size in both configurations. In Stage 2, low-rank adaptation adds 168M LoRA parameters on KriKri-8B and 133M on Racka-4B, while the Stage 1 bridge remains frozen.

3.2. 2-Stage LoRA Curriculum and Training Hyperparameters

In Stage 1, the LLM and Whisper encoder are fully frozen and only the bridge network is optimized, for 45000 steps, with learning rate 10^{-4} , an effective batch size of 64 (per-device batch size 4 with 4 gradient-accumulation steps across 4 GPUs), bf16 mixed precision, and the AdamW [21] optimizer under a cosine scheduler with warmup. In Stage 2, LoRA is activated on the LLM with rank $r = 64$, scaling $\alpha = 2r = 128$, dropout 0.1, targeting the attention projections and the gated-MLP up/down/gate projections [22]; the bridge network trained in Stage 1 (adapter, downsampler, and cross-attention modules) is frozen for the remainder of training so that LoRA adapts the LLM to the now-fixed acoustic interface rather than co-adapting with a still-moving bridge. This second stage is trained for an additional 45,000 steps, maintaining a peak learning rate of 10^{-4} . To reduce memory footprint, the LLM is fine-tuned leveraging QLoRA [23] via 4-bit NormalFloat (NF4) quantization with double quantization.

3.3. Corpora

We evaluate our architecture on diverse speech corpora spanning formal, read, and spontaneous registers for both languages. For Greek, the datasets include Common Voice [24], FLEURS [25], HParl [26], TEDx [27], Logotypographia [28]. For Hungarian, we utilize Common Voice [24], FLEURS [25], Speech Massive [29], VoxPopuli [30], and Yodas [31]. Additionally, the Hungarian training mixture incorporates two commercially licensed corpora (anonymized here as Internal-1 and Internal-2); because these are proprietary, they are utilized exclusively for training and validation and are omitted from the benchmark evaluation tables. Each corpus is split into train/validation/test following a fixed priority: the creators’ own official splits, if provided; otherwise, a leave-subject-out (LSO) split by speaker/subject ID, ensuring no speaker overlaps train and test, using a 60/20/20 ratio. Because dataset sizes vary by orders of magnitude, training batches are constructed via interleaved sampling across corpora (with a temperature factor of $T = 1.2$).

3.4. Preprocessing, Prompting, and Inference

Transcripts are normalized prior to training and evaluation, by removing non-spoken artifacts, such as speaker-tags and transcriber

annotations. The text is further simplified, by standardizing punctuation marks and explicitly verbalizing numerical digits into their spoken word forms. Crucially, punctuation signs are retained rather than stripped; this ensures the model is trained to actively predict punctuation and syntactic boundaries directly from the acoustic signal and linguistic context. Each utterance is additionally conditioned on a natural-language epistemic prompt telling the model what transcription style to produce: [PROMPT_CLEAN] (accurate, clean transcript), [PROMPT_VERBATIM] (word-for-word transcript, disfluencies retained), or [PROMPT_FORMAL] (strictly formal/official transcript). Each corpus is assigned one style matching its own reference-transcription convention, applied consistently at training and inference. During training, Whisper’s input features are also regularized with SpecAugment [32], which helps prevent the network from overfitting. During inference, our model generates transcripts using beam search (beam width of 4) with a repetition penalty of 1.15 and a 512-token maximum constraint.

4. RESULTS AND DISCUSSION

We report our model’s performance across three metrics: standard Word Error Rate (WER), Normalized Word Error Rate (N-WER), and Character Error Rate (CER). N-WER calculates the error rate after stripping punctuation and casing variations, isolating the model’s core phonetic and orthographic accuracy. We utilize N-WER as our primary comparative metric (denoted as Δ_W) to ensure a fair and direct comparison, as it is the standard benchmark metric reported by the VoxKrikri framework [12].

4.1. Greek: Comparison Against Baselines

Table 1 compares against full-context VoxKrikri [12] and Whisper-v3 across five Greek corpora. Our model outperforms VoxKrikri on Common Voice ($\Delta = -4.74$), HParl ($\Delta = -4.81$), and Logotypographia ($\Delta = -2.80$), but FLEURS is the one corpus where VoxKrikri’s simpler single-stage, single-injection design wins ($\Delta = +1.57$), though we still improve over Whisper-v3 on this dataset. We attribute this primarily to FLEURS’s dense foreign named entities, which our LLM’s linguistic prior tends to mis-transliterate (Section 4.4); complementarily, VoxKrikri’s simpler single-injection design may itself suit FLEURS’s short, read-style utterances better, benefiting less from our deeper multi-layer grounding.

Table 1: Greek ASR performance presented as WER / N-WER / CER (%). Δ_W represents the N-WER difference between our model and the best competitive baseline.

Dataset	Ours	Δ_W	Vox (full)	Whisper
Common Voice(34h) [24]	7.3 / X.X / X.X	-4.74	12.04	13.98
FLEURS(12h) [25]	10.9 / X.X / X.X	+1.57	9.33	11.02
HParl(120h) [26]	8.2 / X.X / X.X	-4.81	13.01	16.99
Logotypographia(52h) [28]	6.6 / X.X / X.X	-2.80	9.4	10.86
TEDx(30h) [27]	17.8 / X.X / X.X	–	–	–

4.2. Hungarian: Comparison Against Baselines

Table 2 reports and compares the model’s performance across all five public Hungarian corpora against the original multilingual Whisper large-v3 [1]. We deliberately exclude the Hungarian-specific Whisper-Hu [33] checkpoint, whose model documentation discloses fine-tuning on the training, validation, and *test* splits of VoxPopuli, Common Voice, and Speech-Massive, making comparison on

those benchmarks non-zero-shot and unfair; Our model outperforms Whisper on four of the five comparable corpora (Table 2), with the largest margins on FLEURS (10.2 vs. 37.8, $\Delta = -27.6$) and Speech Massive (38.2 vs. 59.1, $\Delta = -20.9$), and further wins on Common Voice ($\Delta = -6.9$), VoxPopuli ($\Delta = -8.0$). The sole loss is Yodas, where plain Whisper remains stronger (11.5 vs. 8.1, $\Delta = +3.4$). This loss likely reflects reference provenance rather than a real shortfall: Yodas transcripts are pseudo-labeled by faster-whisper-large-v3 itself, so comparing us to zero-shot Whisper on Whisper-generated references is close to self-referential evaluation.

Table 2: Hungarian ASR performance presented as WER / N-WER / CER (%). Δ_W represents the N-WER difference between our model and the original Whisper baseline.

Dataset	Ours	Δ_W	Whisper
Common Voice(180h) [24]	9.0 / X.X / X.X	-6.9	15.9
FLEURS(12h) [25]	10.2 / X.X / X.X	-27.6	37.8
Speech Massive(120h) [29]	38.2 / X.X / X.X	-20.9	59.1
VoxPopuli(120h) [30]	9.8 / X.X / X.X	-8.0	17.8
Yodas(150h) [31]	11.5 / X.X / X.X	+3.4	8.1

4.3. Model Interpretability

To empirically substantiate the depth specialization hypothesized in Section 2.2, we evaluated attention entropy ($\bar{H}^{\text{norm}}$) and Pearson diagonality (r) over 200 random samples from the Common Voice dataset across the three Racka injection layers [15, 25, 35]. As visualized in Fig. 2, our quantitative analysis confirms distinct functional roles for each depth: (i) the earliest layer (15) exhibits high entropy ($\bar{H}^{\text{norm}} = 0.951 \pm 0.008$) and moderate, highly variable diagonality ($r = 0.534 \pm 0.377$), consistent with coarse detection of locally salient acoustic events; (ii) the middle layer (25) achieves sharp temporal focus, characterized by the lowest entropy ($\bar{H}^{\text{norm}} = 0.759 \pm 0.041$) and highest monotonic diagonality ($r = 0.887 \pm 0.109$). Notably, this precise grounding occurs at $\sim 69\%$ relative depth of Racka’s 36 decoder layers, corroborating independent findings for VoxKrikri [12], and (iii) the latest layer (35) shows a relaxation in temporal tracking, with elevated entropy ($\bar{H}^{\text{norm}} = 0.788 \pm 0.027$) and reduced diagonality ($r = 0.704 \pm 0.199$), confirming its role in diffuse, non-monotonic global re-grounding. This quantitative depth specialization strongly supports our persistent-acoustic-memory motivation.

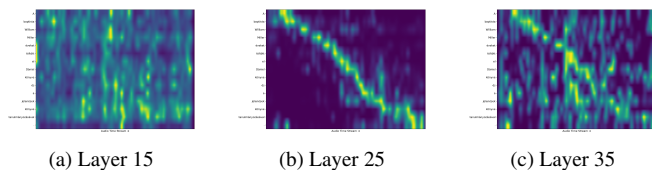


Fig. 2: Word-level cross-attention alignment for a representative Hungarian test utterance, shown separately for each of the three injection layers

4.4. Limitations

Our qualitative analysis surfaces a recurring failure mode: because the fused LLM carries strong linguistic priors, it sometimes acts

less like a verbatim transcriber and more like a “copy editor”. Observed behaviors include silently dropping hesitation markers and dysfluencies, normalizing colloquial contractions to their formal written form, occasionally completing thoughts on abruptly cut-off audio, and mis-transliterating foreign proper names, inflating WER on disfluency-rich, verbatim-style corpora even when the output is fluent and largely faithful in meaning. Our evaluation is also limited to two medium-resource languages, a single frozen acoustic encoder (Whisper-large-v3); generalization to other languages, encoders, or truly low-resource settings remains untested.

4.5. Ablation Study

To isolate the performance contributions of our individual architectural choices, we conduct a systematic ablation study. Table 3 outlines the progressive ablation needed to attribute the reported gains to individual design choices rather than their combination. To circumvent the prohibitive expense of executing multiple full-length training cycles, all ablation metrics are evaluated on the validation set specifically at the 12,000-step training checkpoint.

Table 3: Ablation study for Hungarian on the validation set, presented as WER. Δ_W represents the WER degradation compared to the Full model.

Configuration	WER (%)	Δ_W vs. Full
Full model	21.4	–
Without Prompt Injection	24.7	+3.3
With 2 injection layers (15, 25)	TBD	+X.X
With 2 injection layers (25, 35)	TBD	+X.X
With 1 injection layer (25)	TBD	+X.X
Without layer-wise acoustic fusion	23.2	+1.8
With input-independent layer weights	22.5	+1.1
Without gated cross-attention	22.5	+1.1
Without text perturbation, $p = 0\%$	22.5	+1.1
With text perturbation, $p = 20\%$	22.1	+0.7
With text perturbation, $p = 30\%$	22.5	+1.1
Cross-Attn Downsampling Factor $k = 5$	21.8	+0.4
Cross-Attn Downsampling Factor $k = 1$	21.2	-0.2

5. CONCLUSION & FUTURE WORK

We presented a Continuous Dual-Fusion architecture coupling a frozen Whisper encoder with language-specialized LLMs for medium-resource ASR in Hungarian and Greek, using multi-layer gated cross-attention, layer-wise acoustic pooling, and text perturbation to ground generation in the acoustic signal without discarding linguistic priors. Across nine Hungarian and five Greek corpora, our model reduces WER over standalone Whisper on all but one Hungarian corpus (Table 2) and beats the strongest published baseline, VoxKrikri, on three of four comparable Greek corpora (Table 1).

We plan to: (i) incorporate acoustic data from higher-resource languages (e.g., English, German for Hungarian and English, Cypriot for Greek) to strengthen shared representations while confining foreign-language exposure to the encoder/cross-attention pathway and monitoring for code-switching leakage; (ii) use TTS-synthesized utterances to expose the linguistic side to target-language words absent from our corpora, particularly for morphologically rich Hungarian; and (iii) explore an audio-forecasting

head predicting the next downsampled acoustic embedding from the LLM’s hidden states at injected audio-token positions, which could serve as transcript-free self-supervised pretraining over unlabeled speech, directly addressing the paired-data scarcity motivating this work.

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